



Modelling Financial Markets

Chapter 1: Stylised Facts and Returns



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Acronyms used in this material (1/2)

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- AR: AutoRegressive (model); in event studies: Abnormal Return
- ARCH: AutoRegressive Conditional Heteroskedasticity
- ARCH-LM: ARCH Lagrange Multiplier (test)
- BET-TR: BET Total Return (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAGR: Compound Annual Growth Rate
- CI: Confidence Interval
- CLT: Central Limit Theorem
- CRIX: CRypto IndeX
- DAX: Deutscher Aktienindex (the German stock index)
- EGARCH: Exponential GARCH
- ETF: Exchange-Traded Fund
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GEO: Government Emergency Ordinance
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- JB: Jarque–Bera (test)
- LB: Ljung–Box (test)
- MDD: Maximum DrawDown
- OHLC: Open, High, Low, Close
- PDV: Path-Dependent Volatility

Acronyms used in this material (2/2)

- ▣ **QQ: Quantile–Quantile (plot)**
- ▣ **RMSE: Root Mean Squared Error**
- ▣ **SE: Standard Error**
- ▣ **TSPL: Time-Shifted Power Law (kernel)**
- ▣ **US: United States**
- ▣ **VIX: CBOE Volatility Index**

Chapter Route and What a Stylised Fact Is

Route of this chapter

- Part I: returns, distribution and tails, dependence and its mechanisms
 - Part II: persistence and the limits of inference, market exceptions, OHLC volatility, performance, a case study, one pass of the AI-assisted research loop
 - Technical detail sits in the [Appendix](#) at the end
-
- Stylised facts are the benchmark every model of returns must reproduce: the i.i.d. (independent and identically distributed) Normal-distribution model fails most of them

Definition

- A **stylised fact** is a qualitative statistical regularity of **returns** seen across assets, markets and periods: “returns have heavy tails”, not “kurtosis = 7.3” [Cont \(2001\)](#)
- It is a hypothesis to re-test on each market and frequency [Ratliff-Crain, Van Oort, Koehler & Tivnan \(2025\)](#)
- Key message: returns show little linear autocorrelation, but their **size** is highly predictable

Data Used in This Chapter

Series	Price used	Type, currency	Calendar	Sample	T
S&P 500	daily close (plus OHLC)	price index, USD	US trading days	1990–2026	9 245
BET-TR	daily close	total return, RON	BVB trading days	Sep 2014 – 2026	2 992
Bitcoin	daily close	price, USD	every day, 24/7	Sep 2014 – 2026	4 384
EUR/RON	BNR reference rate	RON per EUR, managed float	Romanian working days	Jul 2005 – 2026	5 352
Gold	daily close	spot, USD per ounce	weekdays only	1990–2026	9 559

- Daily closes up to 18 September 2026; daily log returns on each series' own calendar
- **Managed float:** a market exchange rate with central-bank interventions; the official BNR fixing replaces the EUR/RON market (interbank) quotes, whose lag-1 autocorrelation of -0.4 to -0.7 in several years came from quote errors
- S&P 500 opens before 2007 equal the previous close: range-based estimators start in 2008

Returns: Definitions and Aggregation

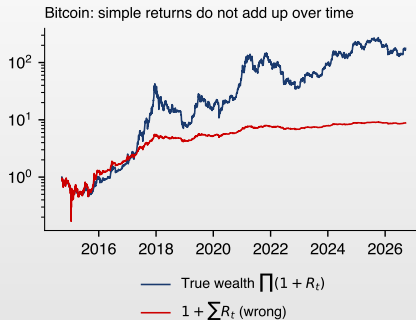
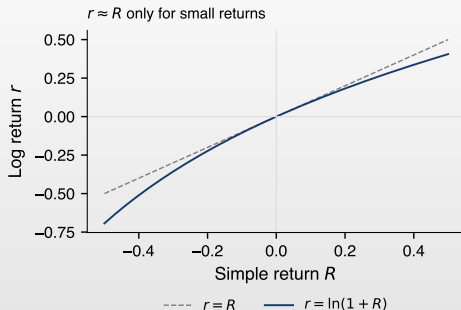
	Simple return R_t	Log return r_t
Definition	$(P_t + D_t)/P_{t-1} - 1$	$\ln(1 + R_t) = \ln P_t - \ln P_{t-1}$ (no dividend)
Over k periods	$\prod_j (1 + R_{t-j}) - 1$	$\sum_j r_{t-j}$: additive over time
Across assets	$R_{p,t} = \sum_j w_j R_{i,t}$: exact	only approximately
Range	$R_t \geq -100\%$	$(-\infty, +\infty)$
Use	portfolios, performance reporting	single-asset dynamics, CLT arguments

- P_t is the close, D_t the dividend paid in $(t-1, t]$; with dividends use total-return or adjusted series (BET-TR)
- $\ln(1 + R) < R$ for every $R \neq 0$: $+1\% \rightarrow 0.995\%$, but $-50\% \rightarrow -69.3\%$
- Textbooks: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 1; [Franke, Härdle & Hafner \(2019\)](#)

Annualisation, CAGR and Volatility Drag

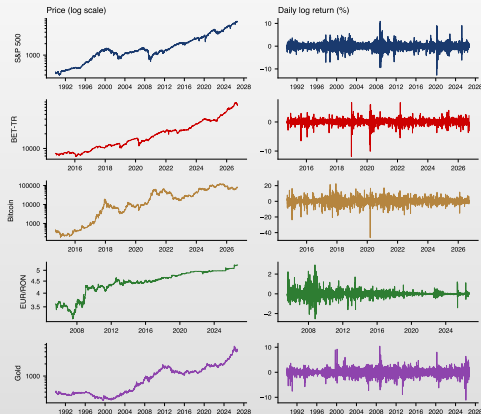
- P = return intervals per year: 252 (stocks), 365 (crypto, traded every day), the actual ≈ 260 weekday quotes (gold)
- $\mu_{\text{ann}} = P \bar{r}$ needs only a constant mean; $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$ is exact only for serially uncorrelated returns with constant variance
 - ▶ with autocorrelation the factor changes: EUR/RON, lag-1 autocorrelation 0.17: 4.4% $\rightarrow$ 5.3% a year (▶ [Appendix](#); Seminar A2)
- **CAGR** (Compound Annual Growth Rate) $= (P_T/P_0)^{1/Y} - 1$, Y = number of years; the geometric mean $\approx \mu_{\text{arith}} - \sigma^2/2$ (**volatility drag**: the gap between the arithmetic and the geometric mean)
 - ▶ S&P 500 1990–2026: $\hat{\sigma} = 1.133\%$ a day, 18.0% a year; CAGR 8.69% (price index, no dividends)
 - ▶ Bitcoin 2014–2026: arithmetic mean 65.2% a year ($P = 365$), CAGR 53.9%: a drag of 11 pp (percentage points)

Simple vs Log Returns in Practice



- Left: $r \approx R$ only for small moves
- Right: Bitcoin 2014–2026 multiplied wealth by 176.9, i.e. $\sum_t r_t = \ln 176.9 = 5.18$; adding daily simple returns gives $1 + \sum_t R_t = 8.83$: wrong by a factor of 20

Five Markets: Prices and Returns



- **Left:** prices on log scale, trending and non-stationary
- **Right:** daily log returns fluctuate around a small mean; calm and turbulent periods alternate; extremes cluster (2008, March 2020)
- EUR/RON moves more than 1% on only 1.4% of days
- Bitcoin, 12 March 2020: log return -46.5% , a price fall of $1 - e^{-0.465} = 37.2\%$

Sample Moments and Their Sampling Error

- ▣ $z_t = (r_t - \bar{r})/\hat{\sigma}$; skewness $\hat{S} = \frac{1}{T} \sum_t z_t^3$, kurtosis $\hat{K} = \frac{1}{T} \sum_t z_t^4$; Normal distribution: $S = 0$, $K = 3$, excess kurtosis $K - 3 = 0$
- ▣ Mean: $SE(\bar{r}) = \sigma/\sqrt{T}$; daily means are 10–60 times smaller than $\hat{\sigma}$, so the mean is the least precise moment
 - S&P 500: $\bar{r} = 0.033\% \pm 0.023\%$ a day (95%), $T = 9,245$
- ▣ i.i.d. data from the Normal distribution: $\sqrt{T} \hat{S} \rightarrow N(0, 6)$ and $\sqrt{T}(\hat{K} - 3) \rightarrow N(0, 24)$
- ▣ Any i.i.d. distribution: the delta method gives a Normal limit for $\hat{S}$ only if $\mathbb{E}r^6 < \infty$, for $\hat{K}$ only if $\mathbb{E}r^8 < \infty$; whether these moments exist is decided by the tail index (Hill estimator, below)

Three Classical Tests and What They Assume

Test	Statistic	H_0	χ^2 limit needs	Problem here
JB, Jarque & Bera (1980)	$T \left[\frac{\hat{S}^2}{6} + \frac{(\hat{K}-3)^2}{24} \right]$	i.i.d., Normal distribution	H_0 itself (all moments exist)	a rejection does not say which part fails
LB, Ljung & Box (1978)	$T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$	$\rho_1 = \dots = \rho_m = 0$	i.i.d., not only uncorrelated	clustering: over-rejects
ARCH-LM, Engle (1982)	TR^2 , $\hat{\varepsilon}_t^2$ on q lags	i.i.d., no ARCH effects	$\mathbb{E}r^4 < \infty$	$\alpha \approx 3 < 4$: size unreliable

- ▣ **ARCH** = AutoRegressive Conditional Heteroskedasticity: today's variance depends on past returns; $\hat{\varepsilon}_t = r_t - \bar{r}$, $q = 5$; α is the tail index (below)
- ▣ We use $m = 10$; 5% critical values: $\chi_2^2 = 5.99$, $\chi_5^2 = 11.1$, $\chi_{10}^2 = 18.3$
- ▣ The question is not how to run the tests but whether their reference distribution holds for returns

Worked Example: JB for the S&P 500

- S&P 500, 1990–2026: $T = 9,245$, $\hat{S} = -0.364$, $\hat{K} - 3 = 10.88$
 - ▶ skewness term $\hat{S}^2/6 = 0.132/6 = 0.022$; kurtosis term $10.88^2/24 = 118.4/24 = 4.93$
 - ▶ $JB = 9,245 \times (0.022 + 4.93) \approx 45,800$, against the critical value 5.99
- Almost all of the statistic comes from the kurtosis term: the problem is the tails, not the asymmetry
- Under the null all moments exist, so the p-value is valid for the **joint** null “i.i.d. and Normal distribution”; heavy tails make JB grow with T : this is power, not a broken p-value
- A rejection does not say which part fails, the marginal distribution or independence; tests of the marginal distribution with HAC variances: [▶ Appendix](#)

Robust Alternative: Quantile-Based Skewness and Kurtosis

- Hinkley skewness $\frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$; Crow–Siddiqui kurtosis $\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$; both 0 for the Normal distribution [Kim & White \(2004\)](#)
- No moments needed; 95% CI from a moving-block bootstrap (blocks of 20 days, 1,000 resamples), valid for stationary, weakly dependent returns with a positive density at these quantiles

	Moment skew.	Hinkley [95% CI]	Excess kurt.	Crow–Siddiqui [95% CI]
S&P 500	−0.36	−0.064 [−0.092, −0.041]	10.9	1.54 [1.29, 1.79]
BET-TR	−1.40	−0.036 [−0.085, 0.025]	17.8	1.06 [0.73, 1.43]
Bitcoin	−0.70	−0.018 [−0.069, 0.032]	12.0	2.25 [1.88, 2.64]
EUR/RON	0.73	0.041 [−0.022, 0.093]	15.1	4.87 [3.74, 6.13]
Gold	−0.23	0.004 [−0.025, 0.030]	9.3	1.47 [1.29, 1.73]

- Excess quantile kurtosis survives in all five markets: no CI contains 0
- Quantile-based negative skewness is established only for the S&P 500; BET-TR's moment skewness −1.40 comes from a few days (19 December 2018)

Summary Statistics: Five Markets

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Observations	9245	2992	4384	5352	9559
Mean (% per day)	0.033	0.077	0.118	0.007	0.025
Std. dev. (% per day)	1.13	0.98	3.49	0.28	1.01
Annualised volatility (%)	18.0	15.5	66.6	4.4	16.3
Skewness	-0.36	-1.40	-0.70	0.73	-0.23
Excess kurtosis	10.9	17.8	12.0	15.1	9.3
Min / max log return (%)	-12.8 / 11.0	-11.9 / 6.7	-46.5 / 22.5	-2.5 / 2.9	-11.1 / 10.4
Jarque-Bera	45805	40433	26482	51575	34732
Ljung-Box $Q(10), r_t$	91.6	70.0	20.2	221.7	32.1
Ljung-Box $Q(10), r_t^2$	8018	714	213	2420	1049
ARCH-LM (5 lags)	2229	371	115	871	483

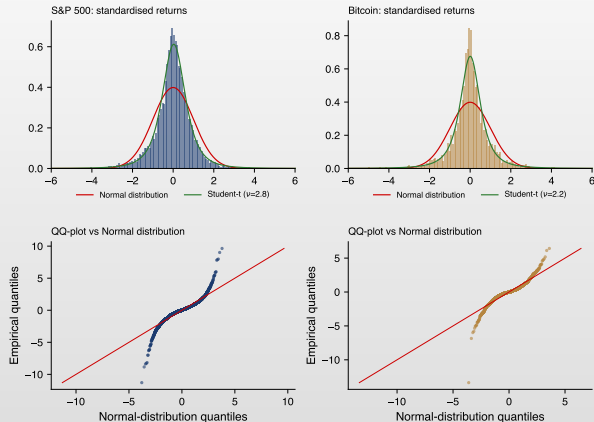
- Descriptive: heavy tails everywhere (excess kurtosis 9.3–17.8); $Q(10)$ on r_t^2 is 10–90 times larger than on r_t : the size of returns is far more predictable than their direction
- Inferential: all i.i.d. p-values are below 0.03, but they are valid only under the assumptions of the previous table

The Stylised Facts of Cont (2001)

Group	Facts
Distribution	2. heavy tails; 3. gain/loss asymmetry; 4. aggregational Gaussianity; 7. conditional heavy tails
Dependence of returns	1. absence of linear autocorrelation
Dependence of volatility	5. intermittency; 6. volatility clustering; 8. slow decay of the autocorrelation of $ r_t $; 11. asymmetry in time scales
Links to other variables	9. leverage effect; 10. volume/volatility correlation

- We re-test them on five markets, with data up to September 2026, adding the inference each test needs
- Early evidence: [Mandelbrot \(1963\)](#), [Fama \(1965\)](#); textbook treatment: [Taylor \(2008\)](#), [Franke, Härdle & Hafner \(2019\)](#)

Fact 2, Heavy Tails: Histograms and QQ-Plots



- Extreme returns are far more frequent than under a Normal distribution with the same variance
- Standardised returns $(r_t - \bar{r})/\hat{\sigma}$: higher peak and fatter tails
- QQ-plots (empirical quantiles against those of the Normal distribution) bend away from the line in both tails
- A Student- t fits much better

How Heavy Are the Tails? Exceedances and the Student- t

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Days beyond $3\hat{\sigma}$ (%)	1.55	1.44	1.82	2.06	1.42
Days beyond $5\hat{\sigma}$ (count)	30	12	10	29	26
Expected under the Normal distribution, $> 5\hat{\sigma}$	0.005	0.002	0.003	0.003	0.005
Student- t degrees of freedom $\hat{\nu}$ (maximum likelihood)	2.8	3.1	2.2	1.2	2.9

- Normal distribution: 0.27% of days beyond 3σ (observed 5–8 times more); one 5σ day every 6,900 years of trading (observed 10–30)
- Student- t , [Praetz \(1972\)](#): $f(x) \propto (1 + \frac{(x-\mu)^2}{\nu s^2})^{-(\nu+1)/2}$; tail $P(X > x) \propto x^{-\nu}$, moments exist only of order $< \nu$; variance $s^2\nu/(\nu - 2)$ for $\nu > 2$, excess kurtosis $6/(\nu - 4)$ for $\nu > 4$
- $\hat{\nu}$ is fitted by **maximum likelihood** to the whole distribution: the many small returns dominate, so a misfit centre moves $\hat{\nu}$; EUR/RON's 1.2 versus a tail-only Hill index of 2.8 signals misspecification (57% of daily changes below 0.1%, plus rare jumps)

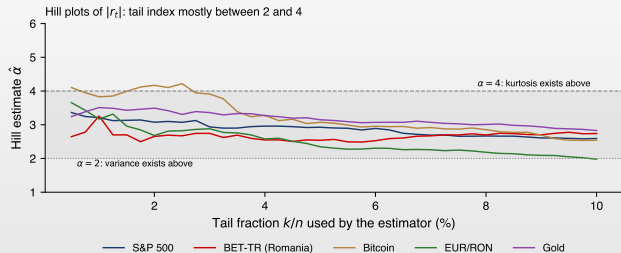
The Hill Estimator of the Tail Index

- **Tail index** α : $P(|X| > x) \approx c x^{-\alpha}$ for large x ; $\mathbb{E}|X|^p < \infty$ only for $p < \alpha$
 - ▶ $\alpha > 2$: finite variance; $\alpha > 4$: finite kurtosis; power laws were first argued by Mandelbrot (1963) and Fama (1965)
- **Hill estimator**, Hill (1975): order $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)}$ and use the k largest

$$\hat{\alpha}_k = \left[\frac{1}{k} \sum_{i=1}^k \ln X_{(i)} - \ln X_{(k+1)} \right]^{-1}, \quad \text{SE}(\hat{\alpha}_k) \approx \hat{\alpha}_k / \sqrt{k}$$

- The SE assumes independent tail observations and a Pareto tail; clustering of extremes makes it larger (▶ Appendix)
- **Choice of k** : small k is noisy, large k lets the centre in and biases $\hat{\alpha}$; read the level where the **Hill plot** ($\hat{\alpha}_k$ against k) is flat, here $k = 1\text{--}5\%$ of n

Hill Plots for Five Markets



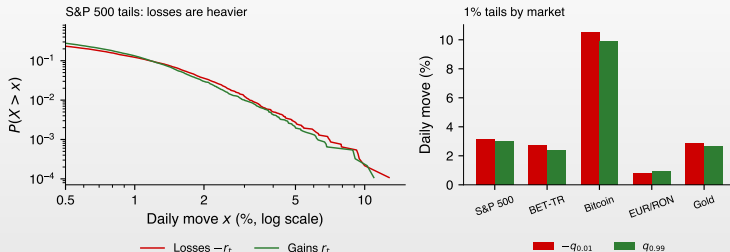
- $\hat{\alpha}$ for $|r_t|$ at $k = 2.5\%$ of n : S&P 500 3.1, BET-TR 2.7, Bitcoin 4.2, EUR/RON 2.8, gold 3.3
- For $k = 1-5\%$: between 2.3 and 3.5 except Bitcoin (3.1–4.2)
- Loss tails heavier for BET-TR (2.2 vs 3.1) and Bitcoin (2.7 vs 3.4)
- Tails are modelled with extreme value theory ([Chapter 7](#))

Tail Indices with Confidence Intervals

$ r_t $, $k = 2.5\%$ of n	n	k	$\hat{\alpha}$	SE i.i.d.	95% CI (i.i.d.)	SE block bootstrap
S&P 500	9245	231	3.07	0.20	[2.68, 3.47]	0.33
BET-TR	2992	74	2.67	0.31	[2.06, 3.28]	0.45
Bitcoin	4384	109	4.22	0.40	[3.43, 5.01]	0.47
EUR/RON	5352	133	2.82	0.24	[2.34, 3.30]	0.28
Gold	9559	238	3.31	0.21	[2.89, 3.73]	0.24

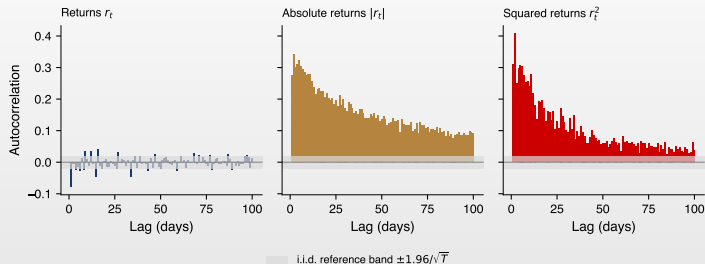
- Moving-block bootstrap (blocks of 20 days, 500 resamples) keeps the clustering of extremes: SEs 1.1–1.6 times the i.i.d. ones
- $\alpha = 4$ is outside every interval except Bitcoin's, also with the bootstrap SE: the fourth moment very likely does not exist
- $\alpha = 2$ is outside every i.i.d. interval; with the bootstrap SE only BET-TR reaches it ($2.67 - 1.96 \times 0.45 = 1.79$)
- Bitcoin's 4.2 uses a threshold where its Hill plot is not yet flat; at a common threshold its two tails give 2.71 and 3.36 ([Appendix](#))

Fact 3: Gain/Loss Asymmetry in the Data



- Survival functions $P(X > x)$ of losses $(-r_t)$ and gains (r_t) , log-log scale: large losses are more frequent than gains of the same size
- **VaR** (Value-at-Risk) 1% = $-q_{0.01}$, the loss exceeded on 1% of days (Chapter 7); it exceeds the 99% return quantile $q_{0.99}$ for 4 of 5 markets, e.g. S&P 500 3.15% vs 2.97%
- EUR/RON is the exception: leu depreciations (up-moves) are larger (0.94% vs 0.76%)
- A Normal-distribution model with the right variance underestimates such quantiles: stress tests based on “ 3σ events” are too optimistic

Facts 1 and 6: Uncorrelated but Not Independent



- Fact 1: $\text{corr}(r_t, r_{t-k}) \approx 0$; S&P 500 1990–2026, lag 1: -0.08 ; grey band: i.i.d. $\pm 1.96/\sqrt{T}$
- Fact 6, volatility clustering: $\text{corr}(|r_t|, |r_{t-k}|) > 0$ for many lags (lag 1: 0.27): large moves follow large moves of either sign
- Independence would require zero correlation of **every** function of returns: returns are uncorrelated but not independent; model $r_t = \sigma_t \varepsilon_t$ with σ_t driven by past returns (GARCH, [Chapter 5](#))

The ACF under Conditional Heteroskedasticity

- ▣ **ACF** (AutoCorrelation Function): $\hat{\rho}_k = \sum_t x_t x_{t-k} / \sum_t x_t^2$, $x_t = r_t - \bar{r}$, plotted against k
- ▣ Uncorrelated returns whose conditional variance moves: $\hat{\rho}_k \approx N(0, \tau_k)$, $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$ [Romano & Thombs \(1996\)](#); i.i.d.: $T\tau_k = 1$; clustering: $T\tau_k > 1$

Lag 1	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
$\hat{\rho}_1$	-0.079	0.061	-0.022	0.170	-0.011
$T\hat{\tau}_1$	5.00	4.35	2.89	6.85	2.43
t -statistic, i.i.d. / robust	-7.6 / -3.4	3.3 / 1.6	-1.5 / -0.9	12.4 / 4.7	-1.1 / -0.7

- ▣ Robust portmanteau $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k$ [Lobato, Nankervis & Savin \(2001\)](#): S&P 500 $Q(10) = 91.6$ but $\tilde{Q}(10) = 19.0$ ($p = 0.041$); its assumptions: [Appendix](#)
- ▣ With robust SEs only the S&P 500 ($t = -3.4$) and EUR/RON ($t = 4.7$) keep a significant lag-1 autocorrelation, economically small

Why Heavy Tails? Mixtures of Normal Distributions

- ▣ Suppose $r_t = \sigma_t \varepsilon_t$ with $\varepsilon_t \sim N(0, 1)$ independent of a random volatility σ_t

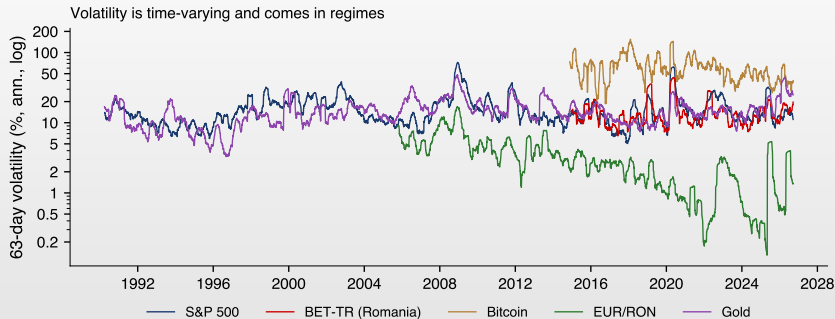
$$K = \frac{\mathbb{E}[r_t^4]}{\mathbb{E}[r_t^2]^2} = 3 \frac{\mathbb{E}[\sigma_t^4]}{\mathbb{E}[\sigma_t^2]^2} \geq 3$$

- ▣ Jensen's inequality ($\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$) with $X = \sigma^2$: any variation in volatility creates excess kurtosis
- ▣ Example: $\sigma = 1\%$ on 90% of days and 3% on 10% of days

$$K = 3 \cdot \frac{0.9 \cdot 1 + 0.1 \cdot 81}{(0.9 + 0.1 \cdot 9)^2} = 3 \cdot \frac{9.0}{3.24} = 8.33$$

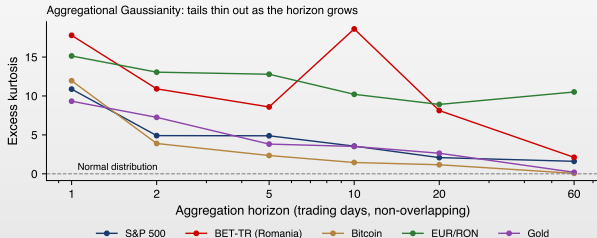
- ▣ This links heavy tails (Fact 2) to volatility clustering (Fact 6); a mixture drawn independently every day has heavy tails but no clustering (Seminar A6, A8)

Fact 5: Intermittency in the Data



- Bursts of high volatility separated by calm periods; constant-volatility models cannot produce them
- 63-day rolling volatility $\sqrt{P} \times$ the standard deviation over the window, log scale; common spikes: 2008–2009, March 2020
- EUR/RON volatility collapsed after 2012: an exchange-rate regime managed by the central bank

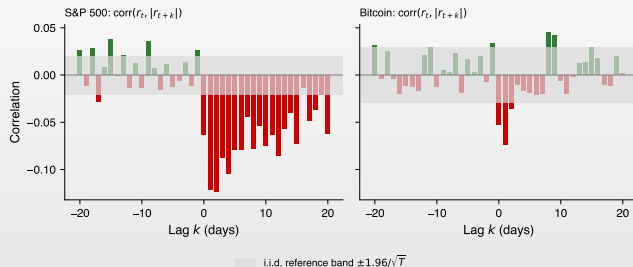
Fact 4: Aggregational Gaussianity



- h -day log return $r_t(h) = r_t + \dots + r_{t-h+1}$ in non-overlapping blocks; the CLT pulls sums towards the Normal distribution, slowly because daily returns are dependent
- Excess kurtosis: S&P 500 $10.9 \rightarrow 1.6$ at $h = 60$, Bitcoin $12.0 \rightarrow 0.1$
- At $h = 60$ only 154 and 73 returns remain: reference SE $\sqrt{24/n} = 0.39$ and 0.57 under i.i.d. normality, so these values are very imprecise
- $h = 60$ is a quarter for the S&P 500 but two months for Bitcoin

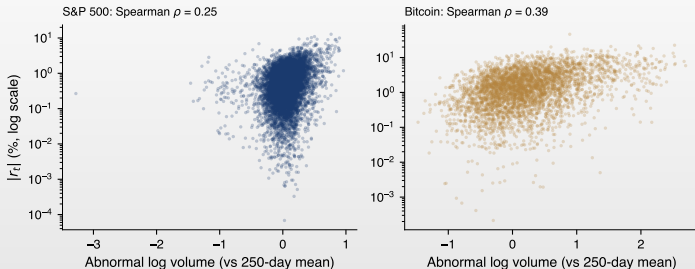
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Fact 9: The Leverage Effect

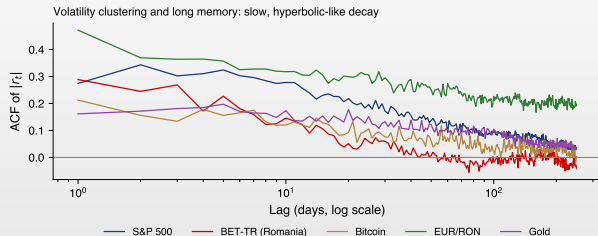


- $L(k) = \text{corr}(r_t, |r_{t+k}|)$, $k > 0$: today's return and a later size; S&P 500: $L(1) = -0.12$, still -0.07 at $k = 10$; no effect for $k < 0$
- BET-TR -0.12 ; weak for Bitcoin (-0.07), negligible for gold (-0.02), reversed for EUR/RON ($+0.10$)
- Possible explanations: financial leverage of firms [Christie \(1982\)](#), the retarded volatility model [Bouchaud, Matacz & Potters \(2001\)](#); models: GJR-GARCH, EGARCH ([Chapter 5](#))

Fact 10: Volume and Volatility Move Together



- Abnormal log volume ($\ln V_t$ minus its 250-day moving average) against $|r_t|$; Spearman rank correlation 0.25 (S&P 500) and 0.39 (Bitcoin)
- Mixture-of-distributions hypothesis [Clark \(1973\)](#): information flow drives both price moves and volume
- Bitcoin volume is exchange-reported and partly **wash trading** (an exchange trading with itself) [Cong, Li, Tang & Yang \(2023\)](#)

Fact 8: Slow Decay of the ACF of $|r_t|$ 

- **Short memory:** $\sum_k |\rho_k| < \infty$, e.g. exponential decay $\rho_k \approx c \phi^k$ (GARCH(1,1), Chapter 5)
- **Long memory:** hyperbolic decay $\rho_k \approx c k^{2d-1}$, $0 < d < \frac{1}{2}$ Ding, Granger & Engle (1993)
- S&P 500: 0.27 at lag 1, 0.22 at lag 20, still 0.09 at lag 100; BET-TR decays fastest
- Slow empirical decay is not yet established long memory: breaks can imitate it (two slides on)

Estimating the Memory Parameter d

- Long memory: spectral density $f(\lambda) \approx G\lambda^{-2d}$ as $\lambda \rightarrow 0$; semiparametric estimators use only the m lowest Fourier frequencies
- Local Whittle Robinson (1995): $\sqrt{m}(\hat{d} - d) \rightarrow N(0, \frac{1}{4})$; log-periodogram regression Geweke & Porter-Hudak (1983): $SE = \pi/\sqrt{24m}$; formulas in the [Appendix](#)

$ r_t , m = T^{0.65}$	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
m	378	181	232	265	386
Local Whittle $\hat{d}$ (SE)	0.48 (0.03)	0.37 (0.04)	0.35 (0.03)	0.39 (0.03)	0.40 (0.03)
Log-periodogram $\hat{d}$ (SE)	0.50 (0.03)	0.41 (0.05)	0.35 (0.04)	0.38 (0.04)	0.42 (0.03)

- $\hat{d} \approx 0.35$ – 0.50 everywhere, at least 8 SEs from $d = 0$ (short memory) with both estimators
- The S&P 500 is close to the stationarity boundary $\frac{1}{2}$

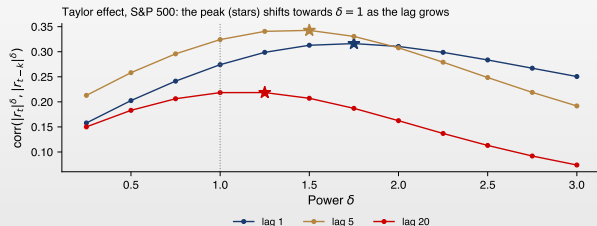
Long Memory or Structural Breaks?

- Rare shifts in the level of volatility produce a slowly decaying ACF of $|r_t|$ and $\hat{d} > 0$ without long memory Diebold & Inoue (2001), Mikosch & Stărică (2004)
- Check: locate one variance break with the cumulative sum of squares Inclán & Tiao (1994), then re-estimate d (local Whittle) on each side

$ r_t $	Break date	Volatility before / after (%)	$\hat{d}$ before (SE)	$\hat{d}$ after (SE)
S&P 500	20 Jul 1998	12.6 / 19.3	0.28 (0.04)	0.51 (0.03)
BET-TR	17 Dec 2018	12.5 / 16.9	0.19 (0.05)	0.37 (0.04)
Bitcoin	19 Jun 2022	74.9 / 47.7	0.39 (0.04)	0.15 (0.05)
EUR/RON	27 Jun 2013	6.6 / 2.2	0.35 (0.04)	0.35 (0.04)
Gold	29 Nov 2005	13.2 / 18.3	0.30 (0.03)	0.37 (0.03)

- $\hat{d}$ stays well above 0 in every sub-sample: one break does not explain the memory
- But $\hat{d}$ moves (Bitcoin $0.39 \rightarrow 0.15$): several breaks or a time-varying d remain open; the break dates are descriptive

The Taylor Effect



- $\text{corr}(|r_t|^\delta, |r_{t-k}|^\delta)$ as a function of the power $\delta > 0$: $\delta = 1$ absolute, $\delta = 2$ squared returns
- S&P 500: maximum at $\delta = 1.75$ (lag 1), 1.5 (lag 5) and 1.25 (lag 20): the optimum moves towards 1 at longer lags and is never at $\delta = 2$
- Documented by [Taylor \(1986\)](#), named by [Granger & Ding \(1995\)](#); motivates power-GARCH models and absolute-value volatility measures

Moments, Tail Index and Valid Inference

Statistic	Needs	With Hill $\hat{\alpha} \approx 2.5\text{--}3.5$
Mean, variance of r_t	$\alpha > 2$	finite, slow convergence
ACF of $ r_t $, robust ACF bands, $\tilde{Q}$	$\alpha > 4$ for Normal limits	doubtful
Kurtosis, ACF of r_t^2 , ARCH-LM, Q on r_t^2	$\alpha > 4$ to exist, $\alpha > 8$ for Normal limits	no standard inference
Skewness	$\alpha > 6$ for a Normal limit	no standard inference
Quantiles, tail counts, Hill $\hat{\alpha}$	no moments	valid, with dependence-robust SEs

- With an infinite fourth moment the sample ACF of r_t^2 converges slowly, to a limit that is not the Normal distribution [Davis & Mikosch \(1998\)](#): the $\pm 1.96/\sqrt{T}$ band and the p-values of Q on r_t^2 and ARCH-LM are not reliable
- The conclusion “volatility clusters” is robust; the exact magnitudes are not; one reason behind the Taylor effect: $|r_t|$ needs fewer moments than r_t^2

Facts 7 and 11: Conditional Tails and Time Scales

Fact 7: conditional heavy tails

- Fit a volatility model, e.g. GARCH (Chapter 5), and standardise: $\hat{z}_t = r_t / \hat{\sigma}_t$
- The kurtosis of $\hat{z}_t$ is lower than that of r_t , still above 3
- Models need heavy-tailed innovations (Student- t , possibly skewed)

- Both facts are documented in Cont (2001); returns standardised by realised volatility from intraday data are close to the Normal distribution Andersen, Bollerslev, Diebold & Ebens (2001) (Chapter 9)

Fact 11: asymmetry in time scales

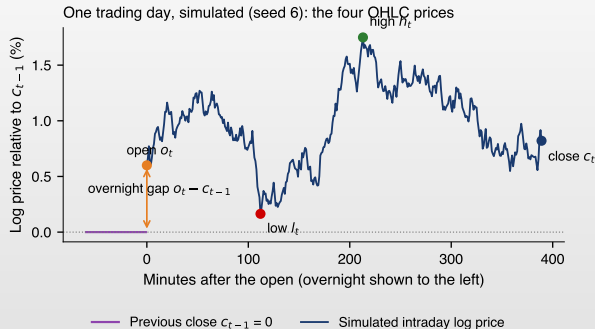
- Weekly volatility predicts daily volatility better than the reverse
- Traders with different horizons react at different frequencies
- **HAR-RV** model (Chapter 9)

Where the Five Markets Differ

Market	Institutional feature	What differs	Candidate explanation
Bitcoin	trades 24/7; $P = 365$	volatility 66.6% vs 18.0%; worst day -46.5% ; weak leverage (-0.07); ACF of $ r_t $ at lag 100: 0.02 vs 0.09	no corporate leverage; volatility rises after large moves of either sign
EUR/RON	BNR daily fixing, managed float	volatility 4.4% but excess kurtosis 15.1; $\hat{\rho}_1 = +0.17$; largest moves October 2008 ($+2.9\%$, -2.5%)	adjustments smoothed over several days
BET-TR	total return, frontier/emerging market	skewness -1.40 ; worst day -11.9% on 19 Dec 2018 (bank-asset tax announced on 18 Dec, GEO 114/2018); $\hat{\rho}_1 = +0.06$	policy shocks; non-synchronous trading of constituents Campbell, Lo & MacKinlay (1997), ch. 3

- The explanations are hypotheses, not identified causes; low volatility is not low tail risk
- Heavy tails and volatility clustering hold in all five; autocorrelation and asymmetry depend on institutions

Why Use Open, High, Low and Close?



- **OHLC**: the Open, High, Low and Close of a day; o_t, h_t, l_t, c_t their logs
- **Close-to-close**: the standard deviation of daily close-to-close log returns; one number per day
- **Range-based** estimators use the whole path: the range $h_t - l_t$ is large on volatile days even if the close equals the open
- **Efficiency**: the ratio of the variances of two estimators

Parkinson and Garman–Klass Estimators

- ▣ **Parkinson**, [Parkinson \(1980\)](#): uses only the daily range

$$\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \frac{1}{n} \sum_{t=1}^n (h_t - l_t)^2$$

- ▣ **Garman–Klass**, [Garman & Klass \(1980\)](#): combines the range with the open-to-close move

$$\hat{\sigma}_{GK}^2 = \frac{1}{n} \sum_{t=1}^n \left[\frac{1}{2} (h_t - l_t)^2 - (2 \ln 2 - 1) (c_t - o_t)^2 \right]$$

- ▣ Derived for a Brownian motion (the continuous-time random walk, [Chapter 11](#)) without drift and without overnight jumps
 - ▶ efficiency relative to close-to-close: about 5 (Parkinson, Seminar A5) and 7–8 (Garman–Klass)

Rogers–Satchell and Yang–Zhang Estimators

- **Rogers–Satchell**, [Rogers & Satchell \(1991\)](#): unbiased under a non-zero **drift** (expected trend of the log price)

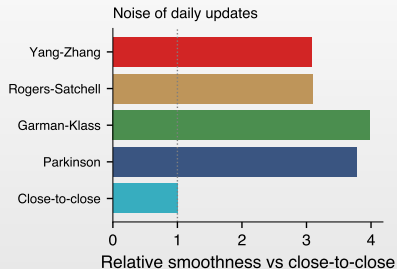
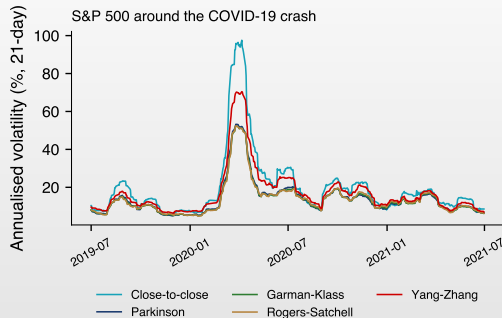
$$\hat{\sigma}_{RS}^2 = \frac{1}{n} \sum_{t=1}^n [(h_t - o_t)(h_t - c_t) + (l_t - o_t)(l_t - c_t)]$$

- **Yang–Zhang**, [Yang & Zhang \(2000\)](#): adds the overnight (close-to-open) jump

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_{\text{overnight}}^2 + k \hat{\sigma}_{\text{open-close}}^2 + (1 - k) \hat{\sigma}_{RS}^2, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

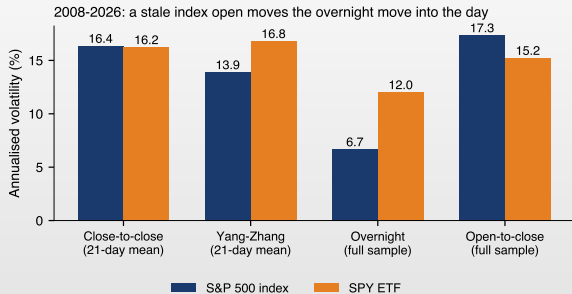
- ▶ $\hat{\sigma}_{\text{overnight}}^2$: variance of $o_t - c_{t-1}$; $\hat{\sigma}_{\text{open-close}}^2$: variance of $c_t - o_t$
- ▶ minimum-variance combination among drift-independent estimators robust to opening jumps
- Range estimators are the bridge to realised volatility from intraday data ([Chapter 9](#))

Estimators on the S&P 500



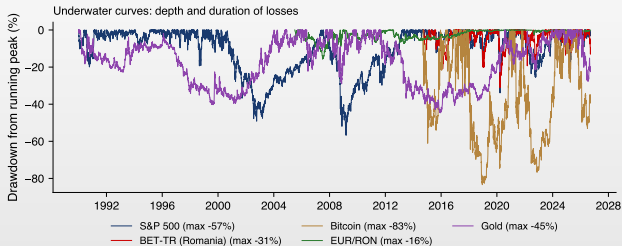
- 21-day windows, 2008–2026; averages: close-to-close 16.4%, Yang-Zhang 13.9%, Parkinson 13.0%, Garman-Klass 12.4%, Rogers-Satchell 12.3%
- Pure range estimators are lower: they miss overnight gaps; the range-based series are 3–4 times smoother, which is a property of the estimates, not by itself proof of efficiency

Stale Opens: the S&P 500 Index versus SPY



- SPY (the ETF that tracks the S&P 500) opens at a traded price; the index open is computed from the constituents' first prices; at 9:30 many stocks have not traded yet, so their previous close enters the open
- Open = previous close: about 96% of days until 2005, 3.5% over 2008–2026 (up to 26% in 2012–2013)
- A stale open moves part of the overnight move into the day: index overnight volatility 6.7% vs 12.0% for SPY, whose open is a traded price
- Yang–Zhang falls below close-to-close on the index only; use a traded instrument for estimators that use the open

Drawdowns Across Markets



- Drawdown $DD_t = P_t / \max_{s \leq t} P_s - 1$;
maximum drawdown $MDD = \min_t DD_t$
(Chapter 0)
- MDD: Bitcoin -83%, S&P 500 -57%, gold -45%, BET-TR -31% (March 2020), EUR/RON -16%
- S&P 500: 9 October 2007 (1,565.15) to 9 March 2009 (676.53): -56.8% needs +131% to recover
- The MDD grows with the sample length:
compare only equal horizons (Appendix)

 MFM_ch1_risk_measures

How Long Does Recovery Take?

	MDD	Peak	Trough	Peak regained
S&P 500	−57%	Oct 2007	Mar 2009	Mar 2013 (5.5 years)
BET-TR	−31%	Jan 2020	Mar 2020	Dec 2020 (0.9 years)
Bitcoin	−83%	Dec 2017	Dec 2018	Nov 2020 (3.0 years)
EUR/RON	−16%	Dec 2005	Jul 2007	Jan 2008 (2.0 years)
Gold	−45%	Sep 2011	Dec 2015	Jul 2020 (8.9 years)

- **Recovery duration:** the time from the peak until the price regains it
- Depth and duration are different risks: gold's drawdown was shallower than Bitcoin's but lasted three times longer
- Fast V-shaped recoveries (BET-TR 2020) versus long underwater periods (gold 2011–2020)
- For EUR/RON a “drawdown” is a fall of the euro against the leu

Sharpe, Sortino and Calmar Ratios

- **Sharpe ratio** Sharpe (1966), Sharpe (1994): $SR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}$ on **simple** returns, $\mu_{\text{ann}} = P\bar{R}$; $r_f = 0$ here, for comparability
 - ▶ S&P 500 1990–2026: $\mu_{\text{ann}} = 9.95\%$ (mean log return 8.33% plus $\approx \sigma^2/2 = 1.62$ pp), $\sigma_{\text{ann}} = 17.97\%$, $SR = 0.55$
- **Sortino ratio** Sortino & van der Meer (1991): $SoR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}^-$, $\sigma^- = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_t, 0)^2}$
 - ▶ the average runs over all T days, positive days counted as 0; S&P 500: $\sigma_{\text{ann}}^- = 12.68\%$, $SoR = 0.78$
- **Calmar ratio** = $\text{CAGR} / |MDD|$; S&P 500: $8.69\% / 56.78\% = 0.15$, driven by one episode

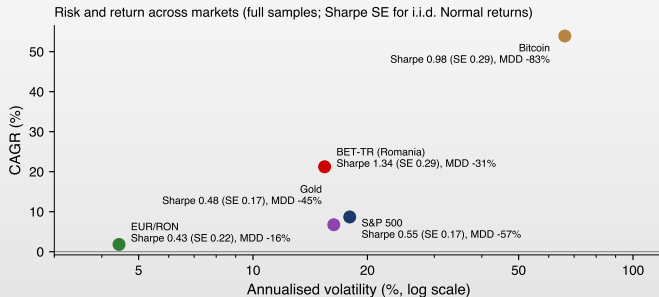
Sharpe Ratios with Standard Errors

- Delta method, i.i.d. returns from the Normal distribution [Lo \(2002\)](#): $\text{Var}(\widehat{SR}_d) \approx (1 + SR_d^2/2)/T$; annualised $SE(\widehat{SR}_{\text{ann}}) \approx \sqrt{P(1 + SR_d^2/2)/T} \approx 1/\sqrt{Y}$: the span matters, not the frequency

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Years Y	36.7	12.0	12.0	21.2	36.7
Sharpe ratio (SE, Normal)	0.55 (0.17)	1.34 (0.29)	0.98 (0.29)	0.43 (0.22)	0.48 (0.17)

- Clustering: HAC or studentised block-bootstrap tests [Ledoit & Wolf \(2008\)](#); they still need finite fourth moments, which the Hill estimates put in doubt
 - ▶ BET-TR vs S&P 500, common days 2014–2026: difference 0.61, $z = 1.75$ ($p = 0.08$), block-bootstrap 95% CI $[-0.01, 1.21]$ (Seminar B13)
- Calmar has no usable sampling theory: its drawdown grows with the sample length

Risk and Return Across Markets



- Sharpe ratios on simple returns, SEs (i.i.d. Normal) in the labels; full table in the [Appendix](#)
- Different sample periods: not a horse race
- S&P 500 without dividends; BET-TR with dividends and in RON, where short-term rates were well above USD rates: with $r_f = 0$ its Sharpe ratio is inflated
- Heavy tails and negative skewness flatter the Sharpe ratio; report it with drawdowns and tail statistics

Case Study (1/5): Question and Target

- Paper: Guyon & Lekeufack (2023), *Quantitative Finance*
- Question: how much of an equity index's volatility can its past daily returns explain, and what function describes this relationship?
 - ▶ target: the 30-day implied volatility index (VIX for the S&P 500) and the 9-day VIX; next-day realised volatility from 5-minute returns (Sections 3.4–3.5 of the paper)
 - ▶ one formula must produce clustering, slow decay and leverage (Facts 6, 8, 9) together; builds on Bollerslev, Hood, Huss & Pedersen (2018)
- Why it matters: if volatility is mostly a function of the price path, it is mostly endogenous; a **path-dependent volatility (PDV)** model needs little extra randomness, unlike stochastic volatility models (Chapters 11–12)
- Data: daily closes of the S&P 500, Euro Stoxx 50, FTSE 100, DAX and Nikkei 225; training 1 Jan 2000 – 31 Dec 2018, test 1 Jan 2019 – 15 May 2022

Case Study (2/5): Model and Features

- Model, eq. (3.7) in the paper, with simple daily returns $r_{t_i} = S_{t_i}/S_{t_{i-1}} - 1$

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \Sigma_t, \quad R_{1,t} = \sum_{t_i \leq t} K_1(t - t_i) r_{t_i}, \quad \Sigma_t = \left(\sum_{t_i \leq t} K_2(t - t_i) r_{t_i}^2 \right)^{1/2}$$

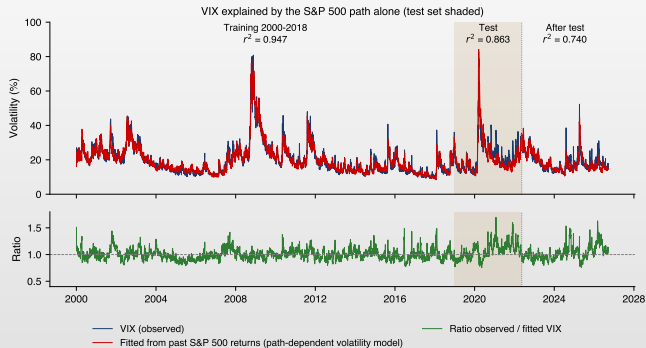
- ▶ σ_t : implied volatility at the close of day t ; $\beta_1 < 0$ gives the leverage effect, $\beta_2 \in (0, 1)$ gives clustering, as in GARCH
- Kernels: **time-shifted power law (TSPL)**, eq. (3.9) in the paper: $K(\tau) \propto (\tau + \delta)^{-\alpha}$, $\alpha > 1$, $\delta > 0$, on 1,000 past days
 - ▶ large weight on the last few days (short memory) and a power-law tail (long memory, Fact 8)
- Seven parameters by nonlinear least squares on the training set; scores: RMSE (root mean squared error) and $r^2 = 1 - \sum(\sigma_t - \hat{\sigma}_t)^2 / \sum(\sigma_t - \bar{\sigma})^2$; estimation details: [▶ Appendix](#)

Case Study (3/5): Published and Replicated Scores

Index	Implied volatility (paper, Table 2)		Realised volatility (paper, Table 4)	
	r^2 train	r^2 test	r^2 train	r^2 test
S&P 500 (VIX)	0.946	0.855	0.738	0.654
S&P 500 (VIX), our replication	0.947	0.863	—	—
S&P 500 (9-day VIX)	0.876	0.914	—	—
Euro Stoxx 50	0.929	0.913	0.672	0.682
Nikkei 225	0.890	0.800	0.563	0.504

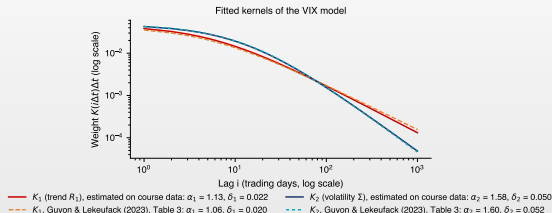
- Past index returns alone explain 80–92% of the variance of implied volatility out of sample; FTSE 100 and DAX in the [Appendix](#)
- On our data (S&P 500 and VIX): RMSE 0.020 training and 0.034 test versus 0.020 and 0.035 in Table 2 of the paper
- Realised volatility scores are lower because 5-minute realised volatility is a noisy target (Remark 3 in the paper); this course uses no 5-minute data, so Table 4 of the paper is not replicated

Case Study (4/5): Observed and Fitted VIX



- Fitted on 3 Jan 2000 – 31 Dec 2018 (4,779 days), started from Table 3 of the paper, then parameters frozen; test set shaded
- After the test set, 16 May 2022 – 18 Sep 2026: $r^2 = 0.740$, RMSE 0.026: a lower RMSE, but the VIX varied less (standard deviation 5.0 versus 9.3 points), so r^2 falls
- Ratio observed/fitted: mean 0.85 in April–May 2020 and 1.16 from October 2020 to May 2022; 8 April 2025: VIX 52.3 vs 39.2 fitted

Case Study (5/5): Fitted Kernels and Limits



- Daily weights of K_2 , log-log scale: $\hat{\alpha}_2 = 1.58$, $\hat{\delta}_2 = 0.050$ versus 1.60, 0.052 in Table 3 of the paper
- The last 5 days carry 20% of the weight, the days beyond lag 250 still 10%: short and long memory in one kernel
- Limits stated by the authors: a small exogenous part remains, the ratio of observed to fitted VIX mean-reverts around 1 and needs a stochastic factor (Section 5 of the paper)
- Open extension: does the same formula describe the realised volatility of Bitcoin or of BET-TR? (project seed)

 MFM_ch1_path_dependent_vol

From Evidence to Model Requirements

Evidence (five markets)	Model requirement	Where in this course
quantile kurtosis > 0 everywhere; Hill $\hat{\alpha} \approx 2.5\text{--}3.5$	heavy-tailed innovations, also conditionally	Ch. 5, Ch. 7 (extreme value theory)
ACF of $ r_t $ positive for months; $\hat{d} \approx 0.35\text{--}0.50$	persistent, time-varying volatility; long memory or breaks	Ch. 5, Ch. 9
$L(1) = -0.12$ (S&P 500, BET-TR); reversed for EUR/RON	asymmetric response to bad news where it exists	Ch. 5 (GJR-GARCH, EGARCH)
robust $\hat{\rho}_1$ economically small except EUR/RON	approximately uncorrelated returns; market microstructure matters	Ch. 2, Ch. 10
quantile skewness negative only for the S&P 500	asymmetry is market-specific	Ch. 7–8

- The random walk with i.i.d. increments from the Normal distribution (Bachelier–Black–Scholes world) violates Facts 2, 3 and 5–11; Fact 4 holds trivially
- ARCH-LM significance supports clustering descriptively; with doubtful fourth moments it is not a formal proof

Open Question: Are Tail Indices Stable over Time?

- The tail index α is usually treated as a constant; crypto-assets changed in structure: futures (2017), spot Bitcoin ETFs (January 2024), institutional holders
 - ▶ [Pele et al. \(2023\)](#): the tail factor separates cryptos from other asset classes
- A typical AI-assistant proposal: “institutional money since the spot ETFs has tamed Bitcoin’s tails”
- Discovery loop in one pass
 - ▶ **literature**: ask for peer-reviewed work with DOIs, then resolve each DOI yourself; [Hoga \(2017\)](#): change-point tests for the tail index valid under β -mixing (e.g. GARCH-type) dependence
 - ▶ **hypotheses**: a change in tail shape, a fall in volatility only (scale does not move α), sampling noise
 - ▶ **data and code**: replicate the chapter’s Hill estimates first (Bitcoin 4.2, S&P 500 3.1)
- Fixed in advance: $k = 2.5\%$ of n , the break date 11 January 2024, block-bootstrap SEs, a placebo asset

Mini-Case: Did Bitcoin's Tail Get Thinner after the ETFs?

- ▣ Test: Hill index of $|r_t|$, $k = 2.5\%$ of n ; $H_0 : \alpha_{\text{post}} = \alpha_{\text{pre}}$ against $H_1 : \alpha_{\text{post}} > \alpha_{\text{pre}}$,
 $z = (\hat{\alpha}_{\text{post}} - \hat{\alpha}_{\text{pre}}) / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$, s from a moving-block bootstrap (blocks of 20 days)
- ▣ Before (Sep 2014 – 10 Jan 2024, $k = 85$): $\hat{\alpha} = 4.16$ ($s = 0.52$); after (11 Jan 2024 – Sep 2026, $k = 24$):
 $\hat{\alpha} = 4.31$ ($s = 0.91$); $z = 0.15$
- ▣ Placebo, S&P 500 at the same date: $2.82 \rightarrow 2.86$, $z = 0.03$
- ▣ **Minimum detectable change**, one-sided 5% test with 80% power: $(1.645 + 0.842) \times 1.05 = 2.6$
 - ▶ a normal approximation, rough with 24 tail observations: the data cannot distinguish “unchanged” from a large thinning of the tail
- ▣ “Not significant” is not “no effect”: wait for more data or pool assets

Your Turn: A Project Seed

- **Question:** did the tail index of Bitcoin and Ethereum change with institutionalisation (futures 2017, spot ETFs 2024)?
 - ▶ raw and GARCH-standardised returns (shape versus scale); loss and gain tails separately
- **Replicate first:** the chapter's Hill table and the mini-case numbers
- **Pre-register:** H_0 , H_1 , the rule for k , break dates and bootstrap, committed before the results
- **Report:** the change-point test of [Hoga \(2017\)](#) as a Quantlet, estimates, both SEs, the minimum detectable change
- **Log:** prompts in `AI_USE.md`, every caught error (an i.i.d. SE, a date chosen from the data) in `AI_ERRORS.md`; reusable prompts and checks in the [▶ Appendix](#)

Key Takeaways and Next: Chapter 2

Takeaways

- ▣ Returns are uncorrelated but not independent: their size is predictable
- ▣ Heavy tails are the rule: tail indices near 3, so the fourth moment very likely does not exist and kurtosis-based inference fails
- ▣ Volatility is persistent (long memory or breaks) and reacts asymmetrically to losses in equity markets
- ▣ Every test needs a reference distribution that the data may violate

Check yourself

- ▣ Why does the $Q(10)$ on Bitcoin returns have $p = 0.028$ but the robust $\tilde{Q}(10)$ $p = 0.29$?
- ▣ Why can the Hill index be estimated when the kurtosis does not exist?
- ▣ **Chapter 2:** are returns predictable at all?

Appendix: Formula Sheet with Assumptions

Returns and annualisation

- ▣ $R_t = P_t/P_{t-1} - 1$, $r_t = \ln(1 + R_t)$, $r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$,
 $R_{p,t} = \sum_i w_i R_{i,t}$
- ▣ $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$
- ▣ $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$: uncorrelated returns, constant variance

Tests

- ▣ $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$: χ_2^2 under i.i.d. Normal
- ▣ $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k)$: χ_m^2 under i.i.d.
- ▣ $\tilde{Q}(m) = \sum_k \hat{\rho}_k^2 / \hat{\tau}_k$: $\mathbb{E}r^4 < \infty$, zero cross terms
- ▣ Hill $\text{SE}(\hat{\alpha}) \approx \hat{\alpha} / \sqrt{k}$: independent Pareto tail

Volatility

- ▣ Parkinson $\frac{1}{4 \ln 2} (\overline{h} - l)^2$: driftless, no overnight jumps
- ▣ Mixture $K = 3 \mathbb{E}[\sigma^4] / \mathbb{E}[\sigma^2]^2$: σ_t independent of ε_t

Performance

- ▣ $SR = (\mu - r_f) / \sigma$ on simple returns; $\text{SE} \approx 1/\sqrt{Y}$ for i.i.d. Normal returns
- ▣ $SoR = (\mu - r_f) / \sigma^-$
- ▣ $DD_t = P_t / \max_{s \leq t} P_s - 1$; Calmar = $\text{CAGR} / |MDD|$

Appendix: Prices Before Computers, the Stock Ticker



Stock ticker designed by Thomas Edison
(Smithsonian collection)

Photo: Jaclyn Nash, Smithsonian National Museum of
American History; public domain; Wikimedia Commons

- Before computers, trades reached brokers as prices printed on a paper **ticker tape**
- A daily close is one snapshot of this stream of prices; returns are built from such snapshots
- The choice of snapshot (close, fixing, a midnight snapshot) matters, as [Chapter 0](#) showed
- Prices at the level of individual trades: market microstructure, [Chapter 10](#)

Appendix: The People Behind the Stylised Facts (1/2)



Carl Friedrich Gauss (1777–1855)

Painting: C.A. Jensen (1840); public domain;
[Wikimedia Commons](#)

- The Normal distribution: the benchmark that returns violate



Louis Bachelier (1870–1946)

Photo: unknown author; public domain;
[Wikimedia Commons](#)

- Prices as a random walk, in his thesis [Bachelier \(1900\)](#)

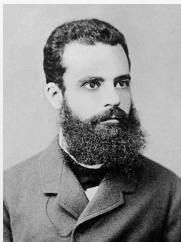


William S. Gosset, "Student" (1876–1937)

Photo: unknown author; public domain;
[Wikimedia Commons](#)

- The Student-t distribution [Student \(1908\)](#), today a model for heavy tails

Appendix: The People Behind the Stylised Facts (2/2)



Vilfredo Pareto (1848–1923)

Photo: unknown author (1870s); public domain; [Wikimedia Commons](#)



Benoît Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 France; [Wikimedia Commons](#)

- Pareto: **power laws** for large incomes, the ancestor of heavy-tail models
- Mandelbrot: cotton prices have heavy tails and the Normal-distribution model fails [Mandelbrot \(1963\)](#)
- Their tail index α is what the Hill estimator in this chapter measures

Appendix: Do Cont's Facts Still Hold? Recent Evidence

- Ratliff-Crain, Van Oort, Koehler & Tivnan (2025)
 - ▶ re-test all 11 facts on intraday returns of the Dow 30 stocks, October 2018 – March 2019
 - ▶ conclusive evidence for 8 facts, no support for the remaining 3
- Heavy tails beyond the Student- t : estimation of alpha-stable distributions, whose tail index is below 2 [Pele \(2014\)](#)
- Entropy-based measures of tail risk as an alternative to kurtosis [Pele, Lazar & Dufour \(2017\)](#)
- Crypto-assets: the same facts, with a tail factor that separates them from other asset classes [Pele et al. \(2023\)](#)

Appendix: Annual Variance under Autocorrelation

- Weakly stationary returns (constant mean and variance, autocovariances depending only on the lag):

$$\text{Var}\left(\sum_{j=1}^h r_{t+j}\right) = h\sigma^2 \left[1 + 2 \sum_{j=1}^{h-1} (1 - j/h)\rho_j\right]$$

- Autoregression of order 1 (AR(1)), $\rho_j = \rho^j$: the factor $\sqrt{VR(h)}$, with $VR(h)$ the variance ratio $\text{Var}(h\text{-day sum})/(h\sigma^2)$: $\sqrt{VR(h)} \rightarrow \sqrt{(1+\rho)/(1-\rho)}$ for large h
 - ▶ EUR/RON, $\rho_1 = 0.17$: factor 1.19, annual volatility 4.4% $\rightarrow$ 5.3%
 - ▶ S&P 500, $\rho_1 = -0.079$: factor 0.92
- Using the sample lag-1 autocorrelation as an AR(1) parameter is an assumption: check the whole ACF (Seminar A2); variance-ratio tests in [Chapter 2](#)

Appendix: What JB Tests, and Tests for Time Series

- JB is a Lagrange Multiplier test [Jarque & Bera \(1980\)](#): under i.i.d. returns from the Normal distribution $\sqrt{T}(\hat{S}, \hat{K} - 3) \rightarrow N(0, \text{diag}(6, 24))$, hence χ_2^2
- GARCH-type dependence changes $\text{Var}(\hat{S})$ and $\text{Var}(\hat{K})$: a rejection mixes the marginal distribution and independence
- [Bai & Ng \(2005\)](#): tests of skewness, kurtosis and normality of the marginal distribution of a time series, studentised with HAC variances
 - ▶ their separate skewness and kurtosis tests (null $S = 0$ or $K = 3$ without normality) need $\mathbb{E}r^6 < \infty$ and $\mathbb{E}r^8 < \infty$

Appendix: The Robust Portmanteau Test and Its Assumptions

- ▣ $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \sim \chi_m^2$ Lobato, Nankervis & Savin (2001) needs $\mathbb{E}r^4 < \infty$ and zero cross terms $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$ for $k \neq l$
- ▣ Dalla, Giraitis & Phillips (2022) keep only the significant cross terms in the covariance of the $\hat{\rho}_k$: none for the S&P 500, BET-TR and Bitcoin
- ▣ $\hat{\tau}_k$ is the variance under $\rho_k = 0$; to compare a non-zero ρ_1 across periods use the HAC SE of the slope of r_t on r_{t-1} (Seminar B6)
- ▣ ACF of $|r_t|$ and of r_t^2 : Normal limits need $\alpha > 4$ and $\alpha > 8$ Davis & Mikosch (1998)

Appendix: Hill Estimator, Asymptotics and Alternatives

- ▣ Hill = maximum likelihood estimator of a Pareto tail above $u = X_{(k+1)}$, conditional on u (Seminar A11)
- ▣ If $k \rightarrow \infty$ and $k/n \rightarrow 0$, and the tail is Pareto up to a second-order term

$$\sqrt{k}(\hat{\alpha}_k - \alpha) \rightarrow N(0, \alpha^2)$$

- ▶ if k grows too fast, a bias term appears: the tail is only approximately Pareto
- ▣ Dependent data (GARCH-type): $\hat{\alpha}$ stays consistent, but extremes cluster and the variance exceeds α^2/k [Hsing \(1991\)](#)
- ▣ Choosing k : plot $\hat{\alpha}$ against $\log k$ ("alt-Hill" plot) [Drees, Resnick & de Haan \(2000\)](#)
- ▣ Regression alternative: log rank on log size, with rank shifted by $-\frac{1}{2}$; $SE = \hat{\alpha}\sqrt{2/k}$ [Gabaix & Ibragimov \(2011\)](#)

Appendix: Two Tails and the Tail of $|r_t|$

- $P(|X| > x) = P(X > x) + P(X < -x)$: with indices α_- (losses) and α_+ (gains), $|X|$ has index $\min(\alpha_-, \alpha_+)$
- At the same k the estimates are not comparable: the k largest $|r_t|$ come from both tails; common threshold u : the lower of the two one-sided thresholds at $k = 2.5\%$ of n

	u (%)	k for $ r_t $	$\hat{\alpha}, r_t $	$\hat{\alpha}_-, \text{losses}$	$\hat{\alpha}_+, \text{gains}$
S&P 500	2.17	508	2.90	2.85	2.96
BET-TR	1.79	153	2.54	2.20	3.06
Bitcoin	7.11	233	3.02	2.71	3.36
EUR/RON	0.54	295	2.28	2.63	2.04
Gold	1.98	513	3.11	2.97	3.28

- At a common u , the $|r_t|$ estimate lies between the two tails; Bitcoin's 4.2 at $k = 2.5\%$ reflects second-order bias: report both one-sided indices

Appendix: Estimating d , the Formulas

- ▣ Fourier frequencies $\lambda_j = 2\pi j/T$, periodogram $I(\lambda_j)$ of the demeaned series; m lowest frequencies, here $m = T^{0.65}$
- ▣ Log-periodogram regression [Geweke & Porter-Hudak \(1983\)](#): regress $\ln I(\lambda_j)$ on $-2 \ln(2 \sin(\lambda_j/2))$; the slope is $\hat{d}$, $SE = \pi/\sqrt{24m}$
- ▣ Local Whittle [Robinson \(1995\)](#):

$$\hat{d} = \arg \min_d \left[\ln \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \ln \lambda_j \right], \quad \sqrt{m}(\hat{d} - d) \rightarrow N(0, \frac{1}{4})$$

- ▣ Local Whittle is robust to non-Gaussianity; a larger m lowers the SE but lets short-run (GARCH) dynamics bias $\hat{d}$ (Seminar B15)

Appendix: Markets in Detail

- ▣ **Crypto-assets:** an asset class with its own statistical profile [Härdle, Harvey & Reule \(2020\)](#); tail, memory and moment factors classify cryptos as a separate asset class, mainly through the tail factor [Pele et al. \(2023\)](#); crypto modelling and the CRIX (CRypto IndeX) [Härdle, Chen & Overbeck \(2017\)](#); Chapter 16
- ▣ **EUR/RON:** BNR reference rate from 1 July 2005, the new leu (RON, 1 RON = 10,000 old lei); long calm periods interrupted by sudden adjustments; EUR/RON market (interbank) quote series showed spurious negative autocorrelation (quote errors)
- ▣ **BET-TR:** the bank-asset tax announced on 18 December 2018 was adopted on 28 December 2018 as Government Emergency Ordinance (GEO) 114/2018; policy shocks produce single-day crashes without comparable rallies
- ▣ **Non-synchronous trading:** index constituents do not all trade at every moment, so the index reflects news with a delay, creating positive autocorrelation [Campbell, Lo & MacKinlay \(1997\)](#), ch. 3

Appendix: The Maximum Drawdown of a Brownian Motion

- Driftless Brownian motion of the log price with annual volatility σ over T years: the maximum **log** drawdown $D = -\ln(1 + MDD) \geq 0$ has

$$\mathbb{E}[D] = \sqrt{\pi/2} \sigma \sqrt{T} \quad \text{Magdon-Ismail, Atiya, Pratap \& Abu-Mostafa (2004)}$$

- $MDD = e^{-D} - 1$; with a positive drift $\mathbb{E}[D]$ grows only like $\ln T$
- Gold 1990–2026: $\sigma = 16.3\%$, $T = 36.7$: benchmark $\mathbb{E}[D] = 1.24$ versus observed 0.592 (−44.7%) (Seminar A4)
- Drawdowns and Calmar ratios from samples of different length are not comparable

Appendix: Sharpe Ratios beyond the Normal Distribution

- i.i.d., any distribution with $\mathbb{E}r^4 < \infty$: $\text{Var}(\widehat{SR}_d) \approx (1 - S SR_d + \frac{K-1}{4} SR_d^2)/T$ [Opdyke \(2007\)](#); here the SEs change by at most 0.02
- The Deflated Sharpe Ratio corrects for non-normality and multiple testing ([Chapter 13](#))

	CAGR	Sharpe	Sortino	Calmar
S&P 500	8.7%	0.55	0.78	0.15
BET-TR	21.2%	1.34	1.87	0.68
Bitcoin	53.9%	0.98	1.45	0.65
EUR/RON	1.8%	0.43	0.65	0.12
Gold	6.7%	0.48	0.69	0.15

 MFM_ch1_risk_measures

Appendix: Case Study, Estimation Details

- Targets (Sections 3.4–3.5 of the paper): the 30-day implied volatility index of each index and the 9-day VIX; realised volatility of day $t + 1$; volatility in decimals ($0.03 = 3$ volatility points)
- Kernels, eq. (3.9) in the paper: $K(\tau) = Z_{\alpha, \delta}^{-1}(\tau + \delta)^{-\alpha}$, normalised so that $\sum_i K(i\Delta t)\Delta t = 1$ on the 1,000-day window, $\Delta t = 1/252$
- Estimation: $(\beta_0, \beta_1, \beta_2, \alpha_1, \delta_1, \alpha_2, \delta_2)$ by nonlinear least squares on the training set, initialised using a lasso regression on features built with exponential kernels
- Our replication: started from Table 3 of the paper; $\hat{\beta}_0 = 0.052$, $\hat{\beta}_1 = -0.090$ (-0.095), $\hat{\beta}_2 = 0.868$ (0.82), $\hat{\alpha}_1 = 1.13$ (1.06); parameter differences alone do not measure precision
- The 9-day VIX exists only from 2011 (shorter training set); the restricted models ($\beta_1 = 0$, $\beta_2 = 0$) are Seminar C3

Appendix: Case Study, Complete Published Results

Index	Implied volatility (paper, Table 2)		Realised volatility (paper, Table 4)	
	r^2 train	r^2 test	r^2 train	r^2 test
S&P 500 (VIX)	0.946	0.855	0.738	0.654
S&P 500 (9-day VIX)	0.876	0.914	–	–
Euro Stoxx 50	0.929	0.913	0.672	0.682
FTSE 100	0.925	0.870	0.650	0.617
DAX	0.934	0.918	0.722	0.557
Nikkei 225	0.890	0.800	0.563	0.504

- VIX parameters: $\beta_0 = 0.057$, $\beta_1 = -0.095$, $\beta_2 = 0.82$; across all targets $\alpha_2 \in [1.2, 2.3]$ (Tables 3 and 5 of the paper): long memory near criticality (Section 3.2 of the paper)
- Dropping Σ_t hurts the most; the model beats two ARCH benchmarks and four published path-dependent models (Figures 3.8–3.9 and Appendix C of the paper)
- Its Markovian 4-factor version, driven by Brownian motion, gives estimated values of about 0.12–0.17 of the Hurst exponent H ($H < 0.5$: a path “rougher” than Brownian motion) for daily realised volatility, as in data (Section 4.6 and Table 9 of the paper): the roughness is only apparent at daily scales

Appendix: The Discovery Loop, Reusable Prompts

- **Literature:** Find peer-reviewed work since 2015 on time variation or structural breaks in tail indices of asset returns, with tests valid under GARCH-type dependence. For each: DOI, estimator, test, data. Flag entries you are unsure of.
- **Hypothesis:** State competing hypotheses for a change in Bitcoin's tail after the 2024 spot ETFs. What does each predict for the Hill index of raw and of GARCH-standardised returns?
- **Data and code:** Use daily Bitcoin closing prices in USD; log returns on its own calendar; Hill index of $|r|$ with $k = 2.5\%$ of n before and after 11 January 2024; moving-block bootstrap standard errors.
- **Robustness:** Write a pre-registered plan: k from 1% to 5% of n , loss and gain tails, GARCH-standardised residuals, block lengths 10 to 50 days, a placebo date and a placebo asset.
- **Critique:** Referee this design. With 24 tail observations after the break, what is the power against a change of 1 in the tail index?

Appendix: What the Researcher Must Check

- ❑ **Wrong return or price series:** reproduce the chapter's Hill table before anything else
- ❑ **Choice of k after seeing the result:** fix the rule ($k = 2.5\%$ of n) in advance; show the Hill plot for 1–5%
- ❑ **i.i.d. standard errors:** $\hat{\alpha}/\sqrt{k}$ ignores volatility clustering; compare with a block bootstrap (for the S&P 500 before 2024 it is 2.2 times larger)
- ❑ **Break date chosen from the data:** use a date fixed by the event, or a sup-type change-point test with its own critical values [Hoga \(2017\)](#)
- ❑ **Bad ticks create fake tails:** list the ten largest $|r_t|$ and match each to a dated event
- ❑ **Low power:** report the minimum detectable change: 2.8 SEs of the difference for a two-sided 5% test with 80% power, 2.5 for a one-sided test

Appendix: Practical Checklist for Empirical Work

- ▣ **Know your data:** price vs total-return index; adjusted vs raw prices; fixing vs traded price; official reference rates where possible (EUR/RON)
- ▣ **Clean before you model:** non-trading days, holidays, stale quotes, obvious errors; match the largest moves to real events
- ▣ **Use log returns** for single-asset dynamics, simple returns for portfolios
- ▣ **Report robust statistics:** quantiles, tail counts and tail indices next to kurtosis
- ▣ **Annualise carefully:** 252 vs 365 periods; the $\sqrt{P}$ rule needs uncorrelated returns with constant variance
- ▣ **Reproducibility:** code and data on GitHub, as Quantlets

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Tukey's g- and h- transformations
- ▣  Applied Time Series Analysis with Python

Code, data and essential reading

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_01
- ▣ Essential reading: Cont (2001); Franke, Härdle & Hafner (2019); Campbell, Lo & MacKinlay (1997), ch. 1–2

References (1/4)

- Andersen, T.G., Bollerslev, T., Diebold, F.X., Ebens, H. (2001). The distribution of realized stock return volatility. *Journal of Financial Economics*, 61(1), 43–76.
- Bachelier, L. (1900). *Théorie de la spéculation*. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- Bai, J., Ng, S. (2005). Tests for skewness, kurtosis, and normality for time series data. *Journal of Business & Economic Statistics*, 23(1), 49–60.
- Bollerslev, T., Hood, B., Huss, J., Pedersen, L.H. (2018). Risk everywhere: modeling and managing volatility. *Review of Financial Studies*, 31(7), 2729–2773.
- Bouchaud, J.-P., Matacz, A., Potters, M. (2001). Leverage effect in financial markets: the retarded volatility model. *Physical Review Letters*, 87(22), 228701.
- Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- Christie, A.A. (1982). The stochastic behavior of common stock variances: value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.
- Clark, P.K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1), 135–155.
- Cong, L.W., Li, X., Tang, K., Yang, Y. (2023). Crypto wash trading. *Management Science*, 69(11), 6427–6454.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Dalla, V., Giraitis, L., Phillips, P.C.B. (2022). Robust tests for white noise and cross-correlation. *Econometric Theory*, 38(5), 913–941.
- Davis, R.A., Mikosch, T. (1998). The sample autocorrelations of heavy-tailed processes with applications to ARCH. *Annals of Statistics*, 26(5), 2049–2080.
- Diebold, F.X., Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159.
- Ding, Z., Granger, C.W.J., Engle, R.F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106.

References (2/4)

- Drees, H., Resnick, S., de Haan, L. (2000). How to make a Hill plot. *Annals of Statistics*, 28(1), 254–274.
- Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Fama, E.F. (1965). The behavior of stock-market prices. *Journal of Business*, 38(1), 34–105.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- Gabaix, X., Ibragimov, R. (2011). Rank – $1/2$: a simple way to improve the OLS estimation of tail exponents. *Journal of Business & Economic Statistics*, 29(1), 24–39.
- Garman, M.B., Klass, M.J. (1980). On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1), 67–78.
- Geweke, J., Porter-Hudak, S. (1983). The estimation and application of long memory time series models. *Journal of Time Series Analysis*, 4(4), 221–238.
- Granger, C.W.J., Ding, Z. (1995). Some properties of absolute return: an alternative measure of risk. *Annales d'Économie et de Statistique*, 40, 67–91.
- Guyon, J., Lekeufack, J. (2023). Volatility is (mostly) path-dependent. *Quantitative Finance*, 23(9), 1221–1258.
- Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance*, 3rd ed. Springer.
- Härdle, W.K., Harvey, C.R., Reule, R.C.G. (2020). Understanding cryptocurrencies. *Journal of Financial Econometrics*, 18(2), 181–208.
- Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- Hoga, Y. (2017). Change point tests for the tail index of β -mixing random variables. *Econometric Theory*, 33(4), 915–954.
- Hsing, T. (1991). On tail index estimation using dependent data. *Annals of Statistics*, 19(3), 1547–1569.

References (3/4)

- Inclán, C., Tiao, G.C. (1994). Use of cumulative sums of squares for retrospective detection of changes of variance. *Journal of the American Statistical Association*, 89(427), 913–923.
- Jarque, C.M., Bera, A.K. (1980). Efficient tests for normality, homoscedasticity and serial independence of regression residuals. *Economics Letters*, 6(3), 255–259.
- Karpoff, J.M. (1987). The relation between price changes and trading volume: a survey. *Journal of Financial and Quantitative Analysis*, 22(1), 109–126.
- Kim, T.-H., White, H. (2004). On more robust estimation of skewness and kurtosis. *Finance Research Letters*, 1(1), 56–73.
- Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- Lobato, I., Nankervis, J.C., Savin, N.E. (2001). Testing for autocorrelation using a modified Box–Pierce Q test. *International Economic Review*, 42(1), 187–205.
- Magdon-Ismail, M., Atiya, A.F., Pratap, A., Abu-Mostafa, Y.S. (2004). On the maximum drawdown of a Brownian motion. *Journal of Applied Probability*, 41(1), 147–161.
- Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.
- Mikosch, T., Stărică, C. (2004). Nonstationarities in financial time series, the long-range dependence, and the IGARCH effects. *Review of Economics and Statistics*, 86(1), 378–390.
- Opdyke, J.D. (2007). Comparing Sharpe ratios: so where are the p-values?. *Journal of Asset Management*, 8(5), 308–336.
- Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.

References (4/4)

- ▣ Pele, D.T. (2014). [A SAS approach for estimating the parameters of an alpha-stable distribution](#). *Procedia Economics and Finance*, 10, 68–77.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). [Information entropy and measures of market risk](#). *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Praetz, P.D. (1972). [The distribution of share price changes](#). *Journal of Business*, 45(1), 49–55.
- ▣ Ratliff-Crain, E., Van Oort, C.M., Koehler, M., Tivnan, B.F. (2025). [Revisiting Cont's stylized facts for modern stock markets](#). *Quantitative Finance*, 25(9), 1343–1373.
- ▣ Robinson, P.M. (1995). [Gaussian semiparametric estimation of long range dependence](#). *Annals of Statistics*, 23(5), 1630–1661.
- ▣ Rogers, L.C.G., Satchell, S.E. (1991). [Estimating variance from high, low and closing prices](#). *Annals of Applied Probability*, 1(4), 504–512.
- ▣ Romano, J.P., Thombs, L.A. (1996). [Inference for autocorrelations under weak assumptions](#). *Journal of the American Statistical Association*, 91(434), 590–600.
- ▣ Sharpe, W.F. (1966). [Mutual fund performance](#). *Journal of Business*, 39(1), 119–138.
- ▣ Sharpe, W.F. (1994). [The Sharpe ratio](#). *Journal of Portfolio Management*, 21(1), 49–58.
- ▣ Sortino, F.A., van der Meer, R. (1991). [Downside risk](#). *Journal of Portfolio Management*, 17(4), 27–31.
- ▣ Student (1908). [The probable error of a mean](#). *Biometrika*, 6(1), 1–25.
- ▣ Taylor, S.J. (1986). *Modelling Financial Time Series*. Wiley; 2nd ed. (2008), World Scientific, doi:10.1142/6578.
- ▣ Yang, D., Zhang, Q. (2000). [Drift-independent volatility estimation based on high, low, open, and close prices](#). *Journal of Business*, 73(3), 477–492.